

Sovereign Cognitive Digital Twins: Fusing 6G ISAC, AI-RAN, and Zero-Trust Edge Grids for Climatic Resilience in Archipelagic and Island Nations

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Abstract

Abstract—Climate-hazard-prone archipelagic and small-island developing states (SIDS) confront an existential risk profile—accelerating sea-level rise, intensifying tropical cyclones, and storm surge—while simultaneously suffering the sparse ground-based instrumentation that makes timely hazard perception difficult. This paper argues for a paradigm shift from passive cellular connectivity to the *Network as a Sensor* (NaaS), realized through a *Sovereign Cognitive Digital Twin* (S-CDT): a federated national digital twin whose perceptual substrate is the 6G radio interface itself. Rather than deploying expensive, disjoint environmental sensing systems, a vulnerable nation can reuse the Integrated Sensing and Communication (ISAC) waveforms of its own network as a distributed radar mesh, feeding a closed-loop cognitive orchestration engine for real-time state estimation, predictive hazard forecasting, and automated mitigation. We ground the architecture in the UK National Digital Twin Programme’s federation lesson and the Gemini Principles; we specify a six-layer S-CDT stack in which ISAC structurally collapses the boundary between the dynamic-data and communication layers; we map the physical layer to the ETSI GR ISC 001 and 3GPP Release 19 sensing frameworks; and we formulate a belief-state control loop—an Extended Kalman Filter feeding a Proximal Policy Optimization agent—that remains stable under O-RAN telemetry delay. Finally, we treat data sovereignty and physical-layer zero-trust security as first-order design constraints, proposing subsystem disaggregation, cross-layer anomaly detection, and privacy-tiered waveforms. Barbados (166 km², ~280k population) is used as a tractable reference deployment, with the Philippines as an archipelagic generalization.

Index Terms—digital twin, 6G, integrated sensing and communication (ISAC), AI-RAN, climate resilience, small island developing states, zero-trust, data sovereignty.

1 Introduction

1.1 The Climate Existential Threat

Archipelagic and small-island developing states occupy the sharp edge of anthropogenic climate change. For nations such as Barbados in the Caribbean and the Philippines in the western Pacific, sea-level rise, coastal erosion, and the intensification of typhoons and storm surge are not distant projections but present-tense determinants of national survival, economic continuity, and population safety [1, 2]. These same nations are, in general, the most weakly instrumented: dense networks of tide gauges, weather radar, and hydrological sensors are capital-intensive to deploy and, critically, fragile precisely during the extreme events for which they are most needed. The result is a perception gap—decisions with hours-to-minutes stakes made on data with hours-to-days latency.

1.2 A Paradigm Shift: The Network as a Sensor

We propose closing that gap not by multiplying dedicated sensors but by re-conceiving infrastructure the nation is already motivated to build. The transition from fifth- to sixth-generation mobile networks introduces *Integrated Sensing and Communication* (ISAC), in which the radio interface that carries data simultaneously performs radar-like sensing of its environ-

ment [7, 9]. Under this *Network as a Sensor* (NaaS) paradigm, every base station is also an environmental instrument, and the coverage footprint of the network becomes the sensing footprint of the nation.

1.3 Thesis and Contributions

Our thesis is that vulnerable nations can deploy a unified *Sovereign Cognitive Digital Twin* (S-CDT) that uses the 6G radio interface as a continuous perceptual nervous system, providing real-time modeling, predictive hazard forecasting, and closed-loop disaster mitigation, under national sovereign control of models and data. This paper contributes: (i) a federated S-CDT reference architecture aligned with established national-digital-twin governance (§2); (ii) a six-layer stack in which ISAC collapses the dynamic-data and communication layers (§3); (iii) a standards-grounded treatment of the physical sensing layer (§4); (iv) a belief-state cognitive-orchestration loop robust to RAN telemetry delay (§5); and (v) a sovereign, physical-layer zero-trust security and privacy design (§6).

2 Grounding the Baseline: The National Digital Twin Paradigm

2.1 The Federated Architecture

The single most consequential design decision is to reject the monolith. The UK National Digital Twin Programme’s central, hard-won lesson is that a national twin is not one model of everything but a *federated ecosystem* of domain-specific twins, connected through shared identifiers, open standards, and a common trust framework [3, 4]. Federation is what makes the system buildable incrementally, governable across institutional owners, and resilient to the failure or replacement of any single component.

2.2 Governance via the Gemini Principles

We structure S-CDT governance around the nine Gemini Principles [4], grouped as:

- **Purpose**—public good, actionable insight, and measurable economic and climatic value.
- **Trust**—security by default (here elevated to *zero trust*), transparent data provenance, and high-quality validation.
- **Function**—federated interoperability, scalability, and long-term evolutionary capability.

For a sovereign deployment the “trust” cluster is load-bearing: the twin’s outputs must be defensible enough to justify evacuations and to underwrite climate finance, which demands auditable provenance rather than best-effort data hygiene.

2.3 The Design Brackets

Two operational national/urban twins bracket the design space. The *built-environment bracket* is Virtual Singapore (Dassault Systèmes with GovTech, SLA, and NRF): a semantically rich 3D model fused with live smart-nation sensor feeds, driving micro-mobility, solar-potential, and flood simulation [5]. The *earth-system bracket* is Destination Earth (DestinE) of the European Commission, with ESA, EUMETSAT, and ECMWF: planetary climate-adaptation and extreme-events twins running on high-performance-computing architectures over a multi-source data lake [6]. The S-CDT merges the brackets—*island-scale built-environment fidelity coupled to earth-system climate dynamics*—and adds the 6G shift that neither exemplar yet exploits: *the connecting network is itself the primary real-time sensor*, redefining the boundary of “live data.”

3 The Unified S-CDT Layered Stack and the ISAC Transformation

3.1 The Core Stack

The S-CDT is organized as six layers over a vertical trust-and-governance spine:

1. **Geospatial Ground Truth**—high-resolution terrain, bathymetry, building envelopes, and critical utility

networks. Barbados’s 166 km² area and ~280k population make complete, high-fidelity national modeling extraordinarily tractable relative to a large state.

2. **Systems of Record**—dynamic registries of land, business, population, and infrastructure assets.
3. **Dynamic Data / Nervous System**—fusion of satellite Earth Observation, IoT telemetry, and the 6G ISAC radio-sensing network.
4. **Simulation and Surrogates**—physics-based hydrology and coastal hydrodynamic models, statistical agent-based evacuation models, and deep-learning surrogate models for real-time ray-tracing and wave propagation.
5. **Synchronization Engine**—reconstruction of current-time state estimates from dynamic physical telemetry.
6. **Interaction and Orchestration**—generative-AI natural-language query interfaces, executive dashboards, and automated control-API pipelines.

3.2 The ISAC Structural Fusion

In the classical layered view, the communication infrastructure is plumbing that *transports* data produced by a physically separate sensing layer. ISAC dissolves this separation: the physical radio waves of the network double as a distributed radar mesh, so the Dynamic-Data Layer and the communication infrastructure become one substrate [8, 9]. Sensor deployment as a distinct capital programme becomes largely obsolete; instead, sensing is a software-and-spectrum function of the network the nation deploys for connectivity. For an operator-integrator this collapses three roles into a single asset—the network is simultaneously (a) physical infrastructure the twin *models*, (b) the backbone that *carries* twin data, and (c) the distributed sensor that *feeds* the twin.

4 The Network as the Perceptive Nervous System: Physical Layer and Standards

4.1 Dual-Function Radar-Communication

At the physical layer, ISAC is realized as Dual-Function Radar-Communication (DFRC): a 6G base station uses shared spectrum and shared hardware to transmit a waveform that simultaneously delivers high-throughput data and yields radar-grade observables—range, angle (angle-of-arrival), radial velocity (Doppler), and micro-structure—by processing echoes and multipath. The ISAC propagation channel is conveniently decomposed as

$$\mathbf{H}_{\text{ISAC}} = \mathbf{H}_{\text{target}} + \mathbf{H}_{\text{background}}, \quad (1)$$

where $\mathbf{H}_{\text{target}}$ captures returns from dynamic sensing targets and $\mathbf{H}_{\text{background}}$ models the quasi-static environment [12, 13]. Estimating and subtracting a learned background is precisely what lets the network isolate a moving hazard—or a weather cell—from clutter.

4.2 Standardization Milestones

ETSI. The ETSI ISG ISAC group’s first report, GR ISC 001 (April 2025), establishes 18 advanced use cases, three integration levels (tight, intermediate, and loose), and six sensing modes for 6G sensing [9]; subsequent reports (e.g. GR ISC 003, 2026) address system and RAN architectures [10]. (*We note a common mis-citation: the 18-use-case report is GR ISC 001, not GR ISC 004.*)

3GPP. Release 19 (study concluded May 2025) specifies wireless sensing use cases in TR 22.837 and extends the TR 38.901 channel model for ISAC [11, 12]. Six sensing modes are defined by transmitter/receiver placement:

- **Monostatic:** TRP-based (Tx/Rx co-located at a transmission-reception point) and UE-based.
- **Bistatic:** TRP→TRP, TRP→UE, UE→TRP, and UE→UE.

Sensing spans FR1, the emerging FR3 mid-band, and FR2 mmWave, trading coverage against range/velocity resolution. Release 20 continues the architecture study.

4.3 Climbing the Sensing Maturity Ladder

Each environmental sub-twin climbs the maturity ladder independently—*Descriptive* → *Diagnostic* → *Predictive* → *Prescriptive* → *Autonomous*—with the predictive-and-above rungs driven natively by edge AI and network sensing. Three exemplar capabilities:

Moving-object detection. Coordinated, multi-static tracking of non-cooperative aerial hazards (unregistered drones over critical infrastructure) and prioritization of emergency vehicles, using Doppler signatures without requiring the target to carry a device.

Environmental and weather mapping. Hyper-local rainfall estimation inferred from frequency-specific attenuation. Specific attenuation follows the ITU-R power law

$$\gamma_R = k R^\alpha \quad [\text{dB/km}], \quad (2)$$

with rain rate R and band-dependent coefficients k, α [14]; the path integral of γ_R is observable as excess attenuation in the channel state, so mmWave/FR3 links act as a dense mesh of virtual rain gauges [15], enabling flash-flood and track-washout prediction well below the resolution of sparse ground radar.

Radio-environment analysis. Continuous RF fingerprinting and multipath-delay-profile indexing dynamically update the local clutter map that constitutes $\mathbf{H}_{\text{background}}$, both improving sensing and providing a change-detection channel in its own right.

5 Closed-Loop Cognitive Orchestration: AI-RAN and Belief-State Control

5.1 AI-RAN Convergence

Perception without timely action is merely telemetry. We distribute intelligence into the Radio Access Network itself: model training via federated learning across sites (keeping raw observations local), and inference at the O-RAN Near-Real-Time and Non-Real-Time RAN Intelligent Controllers (RICs) [16]. The Non-RT RIC hosts the digital-twin models and policy learning; the Near-RT RIC executes low-latency control (beam and resource allocation) as an xApp.

5.2 Belief-State Control under Telemetry Delay

The central control obstacle is delay: O-RAN telemetry can reach the controller tens of milliseconds late (up to ~ 100 ms), so the instantaneous measurement is a stale view of a fast-moving hazard field. We therefore control on a *belief state* rather than raw observations. An Extended Kalman Filter (EKF) running inside the twin maintains an estimate of the latent environmental and network state $\mathbf{x}_t$. The prediction step is

$$\hat{\mathbf{x}}_{t|t-1} = f(\hat{\mathbf{x}}_{t-1|t-1}, \mathbf{u}_t), \quad (3)$$

$$\mathbf{P}_{t|t-1} = \mathbf{F}_t \mathbf{P}_{t-1|t-1} \mathbf{F}_t^\top + \mathbf{Q}_t, \quad (4)$$

with $\mathbf{F}_t = \partial f / \partial \mathbf{x} |_{\hat{\mathbf{x}}_{t-1|t-1}}$. To absorb a measurement delay τ , the update fuses the delayed observation $\mathbf{z}_{t-\tau}$ against the correspondingly retrodicted state,

$$\mathbf{K}_t = \mathbf{P}_{t|t-1} \mathbf{H}_t^\top (\mathbf{H}_t \mathbf{P}_{t|t-1} \mathbf{H}_t^\top + \mathbf{R}_t)^{-1}, \quad (5)$$

$$\hat{\mathbf{x}}_{t|t} = \hat{\mathbf{x}}_{t|t-1} + \mathbf{K}_t (\mathbf{z}_{t-\tau} - h(\hat{\mathbf{x}}_{t-\tau|t-1})). \quad (6)$$

The resulting belief $s_t = (\hat{\mathbf{x}}_{t|t}, \mathbf{P}_{t|t})$ —the mean estimate together with its covariance—is a sufficient statistic that explicitly carries the controller’s uncertainty into the decision.

5.3 Reinforcement-Learned Orchestration

A Proximal Policy Optimization (PPO) agent maps the belief state to control actions a_t —beamforming power, sensing dwell/scan allocation, and mode selection across the six ISAC modes—maximizing the clipped surrogate objective

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min(\rho_t(\theta) \hat{A}_t, \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t) \right], \quad (7)$$

with probability ratio $\rho_t(\theta) = \pi_\theta(a_t | s_t) / \pi_{\theta_{\text{old}}}(a_t | s_t)$ and advantage estimate $\hat{A}_t$. The reward balances the competing objectives of a shared sensing/communication resource,

$$r_t = \alpha \mathcal{A}_{\text{sens}} + \beta \mathcal{T}_{\text{comm}} - \gamma \mathcal{E} - \delta \mathcal{R}_{\text{sec}}, \quad (8)$$

rewarding sensing accuracy $\mathcal{A}_{\text{sens}}$ and communication throughput $\mathcal{T}_{\text{comm}}$ while penalizing energy $\mathcal{E}$ and residual security risk $\mathcal{R}_{\text{sec}}$. Under an active-hazard regime the learned policy shifts resource toward tracking; under nominal conditions it favors throughput.

5.4 Edge Observation Compression

Multi-static sensing produces high-dimensional 3D point clouds that would saturate rate-limited emergency backhaul exactly when links are degraded. We apply an autoencoder (AE) at the gNodeB for dimension reduction: an encoder f_ϕ maps a point cloud $\mathbf{P}$ to a compact code $\mathbf{z} = f_\phi(\mathbf{P})$, transmitted in place of the raw cloud and reconstructed by g_ψ at the core, trained to minimize

$$\mathcal{L}_{\text{AE}} = d_{\text{CD}}(\mathbf{P}, g_\psi(f_\phi(\mathbf{P}))) + \lambda \|\mathbf{z}\|_1, \quad (9)$$

where d_{CD} is the Chamfer distance and the ℓ_1 term encourages a sparse, low-rate code. This preserves hazard-relevant geometry under a strict backhaul budget.

6 Sovereign Data Protection and Physical-Layer Zero Trust

6.1 The Sovereign Compute Grid

A twin that governs evacuations and underwrites climate finance is national critical infrastructure. Climate-vulnerable nations must therefore retain sovereign control of AI weights, models, and spatial telemetry rather than renting perception from external cloud monopolies. The S-CDT is designed for an on-territory sovereign compute grid—edge inference at the RAN, national core for training and archival—so that both the raw sensing of the population and the learned models derived from it remain under domestic jurisdiction. This is also a resilience property: sovereign edge autonomy keeps the twin operating when international connectivity is severed by the very disaster it must manage.

6.2 Zero-Trust Subsystem Disaggregation

Because the network now senses the nation, its compromise is a national-security event, not merely a service outage. We adopt a disaggregated trust model in the spirit of OPPO’s “Minimized Kernel + N Subsystems” 6G architecture [17, 18]: a minimal, formally hardened kernel provides base connectivity, while sensitive functions—national-security sensing, emergency services, and the environmental-sensor plane—run as isolated subsystems with independent trust domains. No plane implicitly trusts another; the environmental-sensing plane cannot be pivoted into from a compromised consumer-data plane.

6.3 Cross-Layer Anomaly Detection

Physical-layer observables are themselves a security sensor. We fuse angle-of-arrival (AoA), Doppler shift, and channel state information (CSI) with higher-layer protocol state to detect physical-layer attacks that are invisible above the PHY: signal spoofing, GNSS/GPS jamming, and tampering with Reconfigurable Intelligent Surfaces (RIS). A spoofed transmitter that is protocol-correct but geometrically impossible (inconsistent AoA/Doppler for its claimed identity) is flagged by the cross-layer detector.

6.4 Privacy-by-Design Waveforms

NaaS sensing is powerful and politically sensitive: the same channel that maps topography can, at fine resolution, read human behavior. We organize spatial telemetry into privacy tiers—**L1 Environmental** (terrain, hydrology, weather), **L2 Behavioral** (aggregate mobility, presence), and **L3 Physiological** (fine micro-Doppler, e.g. respiration/gait)—with escalating consent and access controls. To enforce the boundary at the physical layer we inject calibrated noise into the reported CSI,

$$\tilde{\mathbf{H}} = \mathbf{H} + \mathbf{N}, \quad \mathbf{N} \sim \mathcal{P}(\sigma_{\text{beh}}), \quad (10)$$

where $\mathcal{P}$ is designed to destroy the high-frequency micro-Doppler components that carry behavioral/physiological signatures while preserving the low-frequency structural returns needed for topography and weather sensing. The network can thus deliver high-resolution environmental perception at L1 while being provably unable to export L2/L3 signatures without explicit, tier-appropriate authorization.

7 Deployment Blueprint: Barbados and the Archipelagic Generalization

We propose a *minimum viable twin* that proves the full vertical slice on a single existential question before federating outward. Its components: (1) a national basemap bootstrapped from Earth-observation foundation models; (2) at least one network-derived live feed—microwave-link/CSI rainfall—designed ISAC-ready, alongside a tide/weather feed; (3) one predictive model (coastal inundation under a parametric storm scenario) driven by the live rainfall field; (4) a generative natural-language query interface for decision-makers; and (5) node and sensor attestation plus data provenance on a sovereign ledger to make outputs finance- and evacuation-grade. Federation then proceeds domain by domain—economy and tourism next, coupled to the climate twin through network-derived mobility—while every newly deployed base station is specified ISAC-ready so the sensing mesh densifies as a byproduct of connectivity roll-out. Barbados serves as a bounded, high-fidelity reference; the Philippines generalizes the design to a multi-thousand-island archipelago where inter-island non-terrestrial (satellite) ISAC and edge autonomy become decisive.

8 Implementation: the Barbados Digital Twin as built

The preceding sections describe the target S-CDT. This section reports what has actually been built to date on the Barbados deployment, inside the Ulap SCOPE application, and separates shipped capability (**[B]uilt**) from designed-but-not-yet-enforced capability (**[D]esign**) so the architecture is not over-claimed. Measured against the sensing-maturity ladder of §4, the deployed twin sits at the *descriptive/diagnostic* rungs: it renders the nation and its exposure and joins hazard return-

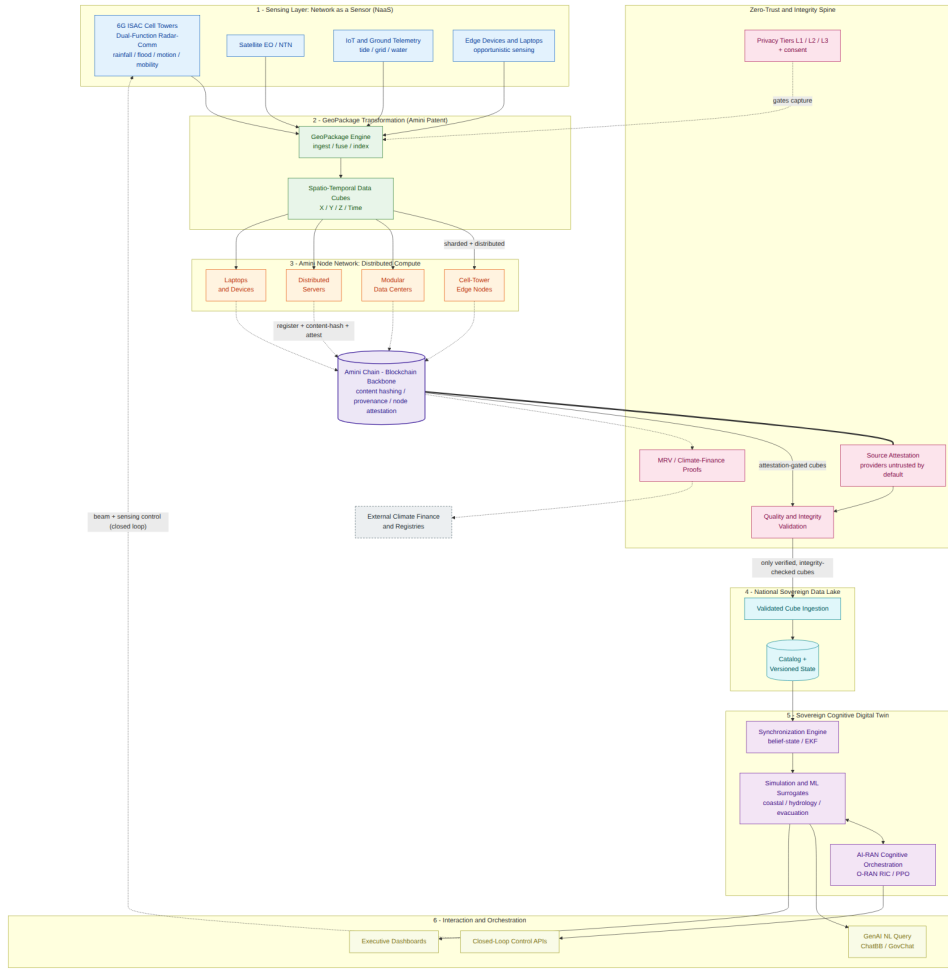


Figure 1: The S-CDT reference architecture referenced throughout this paper: a six-layer stack over a vertical zero-trust and integrity spine. Sensing (Network-as-a-Sensor) feeds the GeoPackage engine that produces spatio-temporal data cubes; cubes are sharded across the Amini node network and anchored on Amini Chain (content hashing, provenance, node attestation); a zero-trust integrity gate admits only verified cubes to the national sovereign data lake that powers the cognitive twin; orchestration closes the loop back to the sensing layer.

periods to assets, but does not yet run the predictive-and-above cognitive loop.

8.1 Mapping the deployment to the reference stack

Figure 1 is the reference architecture; Table 1 maps each of its layers to the deployed system. The build faithfully realizes the *perceptual and serving* substrate — layers 1, 2, 4 (catalog) and 6 (dashboards) — while the *cognitive* (5), *ledgered-trust* (3) and *enforced-governance* (spine) elements remain specified and scheduled (§8.7).

8.2 Data provenance: authoritative layers

All 19 static layers derive from a single authoritative source, the Barbados Geoportal of the Lands & Surveys Department, received as 19 ESRI shapefiles constituting a national infrastructure, hazard and vulnerability database [19]. Two

facts govern every transformation. First, source geometry is EPSG:21292 (Barbados 1938 / British West Indies Grid) and is reprojected to EPSG:4326 for web mapping; a correctness invariant is that the building bounding box must reproject to the whole island (verified at $[-59.650, 13.045]$ to $[-59.422, 13.335]$). Second, critical-asset layers carry hazard return-period fields (rainfall and coastal flood, landslide, seismic), which is what lets the twin join assets to the climate-risk models. Table 2 is the ingested inventory. The government `vulnerability_*` grid is treated as authoritative and supersedes the earlier census-derived parish proxy; the service degrades to the lower-provenance proxy only when the grid is absent, and carries the source label either way.

8.3 Live feeds

Beyond the static export, typed connectors hydrate live layers, one documented ingress per source: 3D-PAWS weather stations via CHORDS, Open-Meteo, OpenSky flights, CelesTrak

Table 1: Reference-stack layers mapped to the Ulap SCOPE deployment. [B] = built, [D] = design.

S-CDT layer	Deployed in Ulap SCOPE today	State
1. Sensing (NaaS)	Live connectors: 3D-PAWS/CHORDS weather, OpenSky flights, CelesTrak TLEs, Sentinel-2/Copernicus EO, Open-Meteo, OpenCellID. ISAC radar-sensing is not yet a feed.	[B] conv. sensing; [D] ISAC
2. GeoPackage transform	<code>twin_ingest</code> : shapefiles → reproject → normalize → GeoPackage + PMTiles — the GeoPackage engine at national scale.	[B]; [D] X/Y/Z/T cubes
3. Node network + chain	Per-layer source attribution + <code>catalog.json</code> manifest. No on-chain content-hashing or node attestation yet.	[D]
4. Sovereign data lake	<code>catalog.json</code> catalog served from a single GeoPackage on sovereign/edge storage; versioning by re-ingest.	[B] catalog; [D] versioned lake
5. Cognitive twin	Sionna RF simulation and climate-risk models exist as surrogates; no EKF belief-state loop or PPO orchestration yet.	[D]
6. Interaction	SCOPE web app: 3D map, twin KPIs, inspect panels, layer/hazard dashboards.	[B]; [D] closed-loop / GovChat
Zero-trust spine	Source attribution on every response + graceful-degrade contract. Tiering, attestation gate and integrity validation designed, not enforced.	[D]

Table 2: Ingested static layer inventory (real counts from `catalog.json`). Source: Barbados Geoportal, Lands & Surveys Dept.; CRS EPSG:21292 → EPSG:4326. `track` = serving strategy.

Layer (twin name)	Geom	Count	Track	Twin role
buildings	polygon	130,248	tiles	Island-wide 3D on terrain
roads	line	705	geojson	Road network
bridges	line	25	geojson	Bridges (hazard-annotated)
ports	polygon	4	geojson	Ports & BGI airport hotspots
antennas	point	100	geojson	Telecom / RF twin
drinking_water_network	line	17,802	tiles	Potable mains
water_mains	line	17,966	tiles	Water mains
wastewater_plants	point	48	geojson	Wastewater plants
desalination_plants	point	5	geojson	Desalination (critical)
dams	point	39	geojson	Dams
reservoirs	point	34	geojson	Reservoirs
pumping_stations	point	19	geojson	Pumping stations
communal_wells	point	38	geojson	Wells
individual_wells	point	40	geojson	Wells
major_projects	point	111	geojson	Capital projects
vulnerability_global	polygon	13,029	tiles	Authoritative gov. vulnerability grid
vulnerability_population	polygon	13,029	tiles	Population exposure
vulnerability_environment	polygon	13,029	tiles	Environmental exposure
vulnerability_equipment	polygon	13,029	tiles	Equipment / economic exposure

TLEs (NTN constellations), Sentinel-2 L2A and Copernicus DEM via the CDSE STAC, ESA WorldCover, Microsoft Global Building Footprints (fallback only), Hansen forest change, Gefabrik/OSM, OpenCellID, TeleGeography submarine cables, and PeeringDB. A strict provenance rule governs overlap: the government export always wins where both exist (for example, government building footprints over Microsoft's), and live feeds serve enrichment, hotspots and layers the export does not cover.

8.4 Zero-trust governance: target versus enforced

The governance model follows §6 and the Gemini Principles: sources untrusted by default, provenance mandatory rather than best-effort, tiered privacy, and sovereign custody. Table 3 states honestly what is enforced today against the target. Source attribution, structural integrity validation (CRS enforcement, null and degenerate-geometry drops, non-finite sanitization) and labeled graceful degradation are shipped; cryptographic content-hashing and on-chain node attestation are the first governance work item on the roadmap. Today the twin serves only L1 (environmental) data, so no L2/L3 export path yet exists to

gate — the privacy tiering is the schema the future ISAC feed will be born into.

8.5 Reproducibility

The served store is fully regenerable from the committed shapefiles by a single idempotent pipeline (`twin_ingest`): per layer it assumes/reprojects to EPSG:4326, drops null and degenerate geometry, applies a declarative rename/keep column spec, writes one table per layer into a GeoPackage (`barbados_twin.gpkg`), emits PMTiles for high-count layers via `tippecanoe`, and writes a `catalog.json` manifest (source, CRS, per-layer count and bounding box). The pipeline deletes and rebuilds the GeoPackage each run, so a new government shapefile drop is absorbed by re-running with no manual merge. The manifest is the machine-readable provenance ledger; the Phase 1 work item extends each entry with a SHA-256 content hash, ingest timestamp, license and privacy tier, turning it into a signed ledger with no change to the serving contract.

Table 3: Zero-trust integrity gate: target versus what is enforced today.

Control	Today in code	Gap to target
Source attribution	✓ source on every FeatureCollection + catalog.json	—
Content hashing	pending	SHA-256 per cube at ingest, on-ledger
Node attestation	pending	Requires Amini Chain backbone
Integrity validation gate	Structural only (finite-geometry checks, null drops, CRS validation)	No cryptographic admission gate
Graceful degradation	✓ gov-grid → census proxy, labeled; empty-but-valid catalog	—
Sovereign custody	✓ single-node, self-hosted; no cloud dependency for the served store	Formalize

8.6 Deployed system

The running Ulap SCOPE twin renders: the island-wide infrastructure network (130,248 buildings, roads, water mains, telecom, ports) draped on 3D terrain; the same scene in oblique 3D at building fidelity; the authoritative per-parish vulnerability index as a graded choropleth with a live weather/KPI header; a government-asset inspect panel (for example Grantley Adams / GAIA International Airport) that surfaces each feature’s source and attributes; and an RF coverage-planning surface driven by the Sionna propagation engine (receiver placement, ray tracing, and RSS/SINR coverage statistics). These views exercise the built layers 1, 2, 4-catalog and 6 of Table 1; screenshots accompany the deployment record.

8.7 Roadmap to the Sovereign Cognitive Digital Twin

The sensing-maturity ladder of §4 is the yardstick, and the roadmap climbs it while closing the [D] gaps of Table 1. **Phase 0 (built)** is the ingest-and-serve substrate documented above. **Phase 1 (next)** closes the highest-value governance gap: content-hash every cube at ingest, stand up the Amini Chain backbone and register the hashes, and promote the structural checks into an explicit quality-and-integrity admission gate so outputs become finance/evacuation-grade. **Phase 2** replaces the single GeoPackage with a versioned lakehouse and lifts layers into X/Y/Z/Time cubes for state history and change detection. **Phase 3** adds the cognitive layer — the EKF belief-state synchronization of §5 and ML surrogates (coastal, hydrology, evacuation) wired to the live rainfall fields. **Phase 4** adds closed-loop PPO/AI-RAN orchestration (an O-RAN Near-RT RIC xApp) and the GenAI GovChat query interface. **Phase 5** brings the 6G ISAC feed online and enforces the L1/L2/L3 privacy waveforms and cross-layer anomaly detection at the physical layer. The methodology is written to transfer: swapping the government geo-portal source and re-running the same ingest/governance pipeline generalizes the build from Barbados to the Philippines archipelago.

9 Conclusion and Future Work

We have argued that the cheapest path to national-scale climate perception for island and archipelagic states is not more dedicated sensors but a re-conception of the network itself as the na-

tion’s sensing nervous system, orchestrated by a sovereign, cognitively closed-loop digital twin. The S-CDT unifies the built-environment and earth-system digital-twin traditions, grounds its perception in the emerging ETSI and 3GPP ISAC standards, controls under realistic RAN delay via belief-state estimation, and treats sovereignty and physical-layer privacy as primary. Future work includes: field quantification of CSI-based rainfall accuracy against gauge networks in the Caribbean band plan; sim-to-real transfer of the PPO orchestration policy on an O-RAN testbed; formal verification of the minimized-kernel trust boundaries; and a measurement study of the L1/L2/L3 privacy-waveform trade-off curve.

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